FIREFLY

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PURPOSE

FireFly addresses the lack of real-time wildfire monitoring in remote, high-risk areas where fixed sensors and cameras may not exist. Its purpose is to give responders earlier, safer awareness of possible ignition points by collecting aerial visual and telemetry data, locating detected fires, and sending actionable alerts to a ground station before conditions escalate.

SOLUTION

FireFly combines ESP32-based drone telemetry, computer-vision fire detection, and a live ground-station map. Drone nodes report GPS position, altitude, and fire-detection status. A YOLO-based vision pipeline detects sustained fire or smoke in video. When multiple drones observe the same event, the ground station triangulates their bearings to estimate the fire's GPS coordinates and displays drone positions, fire markers, and alerts for the operator.

OBJECTIVES

- Detect sustained fire or smoke from aerial video
- Send drone GPS and fire-status telemetry to a ground station
- Triangulate fire location from 2+ drone observations
- Display live drone positions, fire alerts, and estimated fire coordinates on a map
- Validate the end-to-end pipeline through simulation and unit tests

TARGET USERS

- Wildfire operations coordinators who need fast, clear situational awareness
- Fire-response and forestry field crews operating in dangerous or hard-to-access terrain
- Environmental researchers collecting fire-risk and field telemetry data
- Future engineering teams extending the prototype into full flight hardware and field deployment

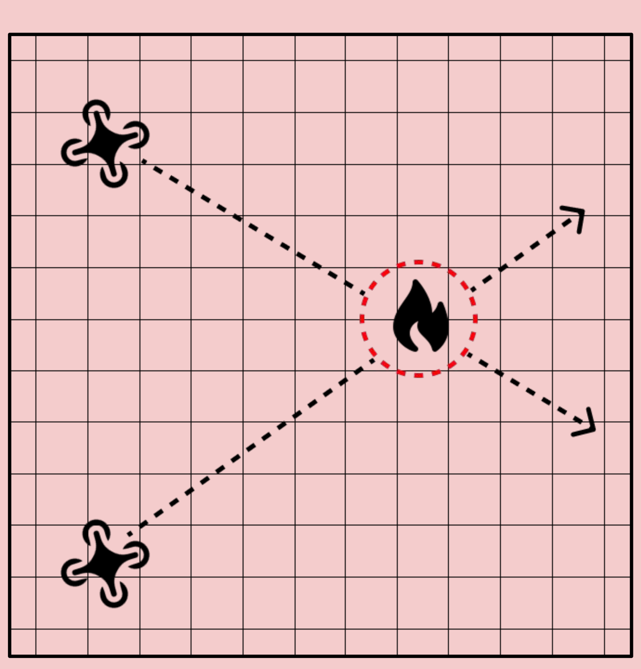
SOFTWARE

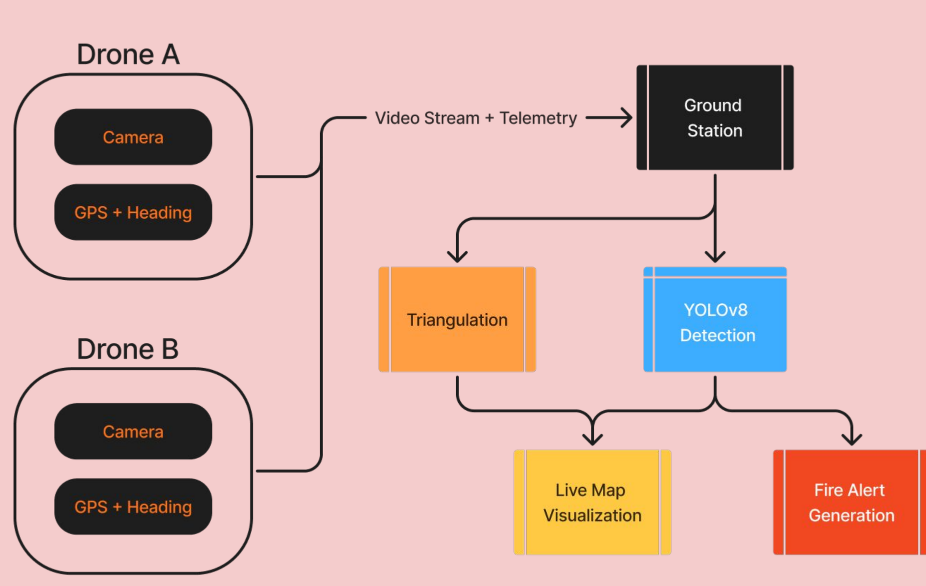
Ground Station Interface
- Live Javascript/Leaflet map visualization
- Displays drone positions, alerts, and estimated fire markers
- Supports simulated fleet testing and local camera discovery

Vision Pipeline
- YOLOv8 fire/smoke detection
- OpenCV video-frame processing
- Sustained-detection filtering to reduce false positives

Localization
- Bearing-based triangulation from 2+ drone observations
- Estimates GPS coordinates of detected fire events
- Unit-tested triangulation calculations

Telemetry Backend
- Flask server receives drone GPS, heading, altitude, and fire status
- Parses real-time ESP32 telemetry messages
- Updates active drone and fire-event state





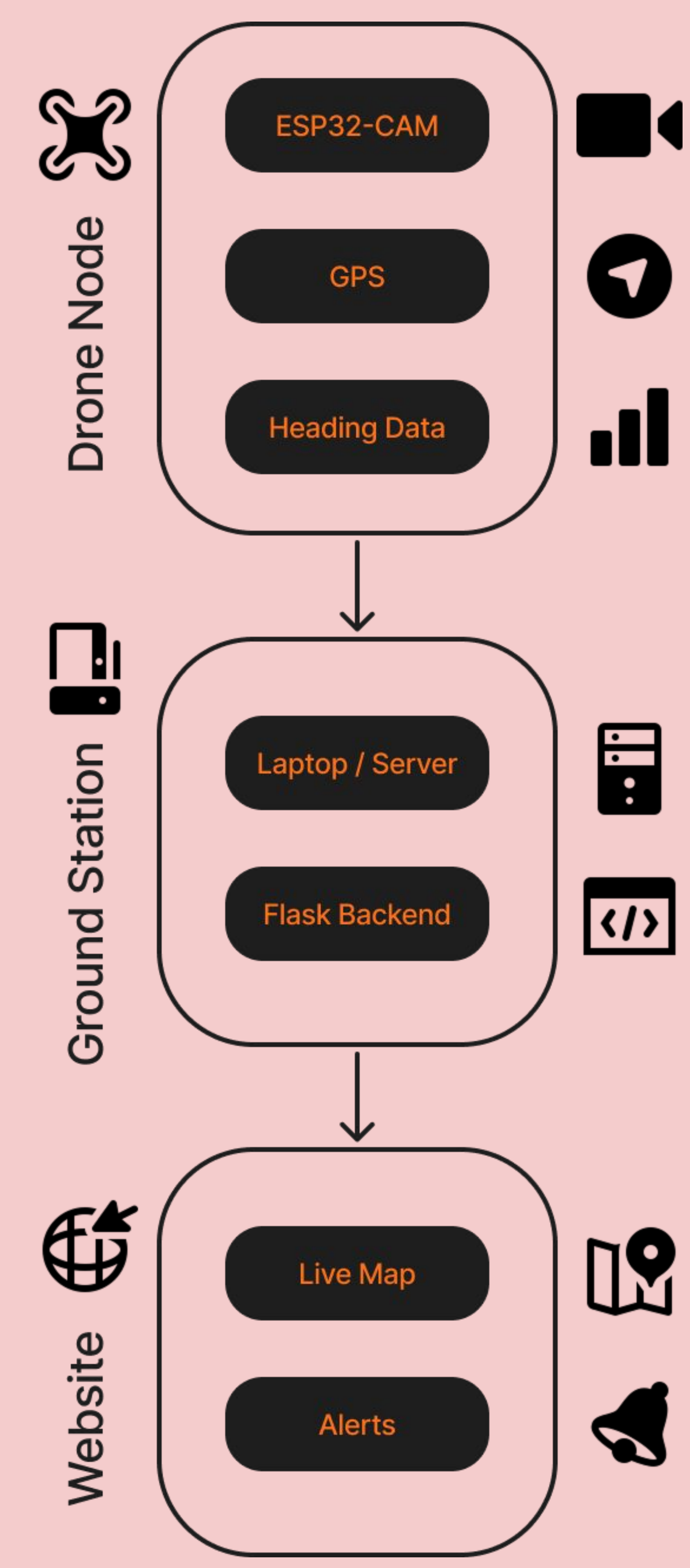
HARDWARE

Drone Node
- ESP32 flight/telemetry microcontroller
- ESP32-CAM video module
- GPS receiver for position data
- Heading/bearing data source

Wireless Communication
- ESP-NOW broadcast telemetry between drone nodes
- Wi-Fi connection to ground station
- Real-time GPS and fire-status transmission

Ground Station
- Laptop/server running Flask backend
- Receives telemetry and fire alerts
- Displays live map and detection results

Sensor Data Collected
- Camera stream for fire/smoke detection
- GPS coordinates
- Drone heading/bearing
- Altitude and telemetry status



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[2] Espressif Systems, "ESP-NOW Programming Guide," docs.espressif.com.
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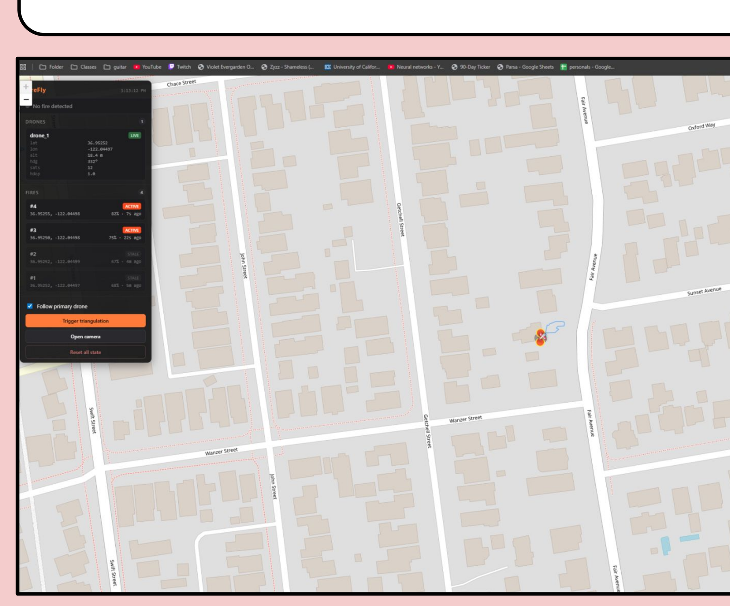
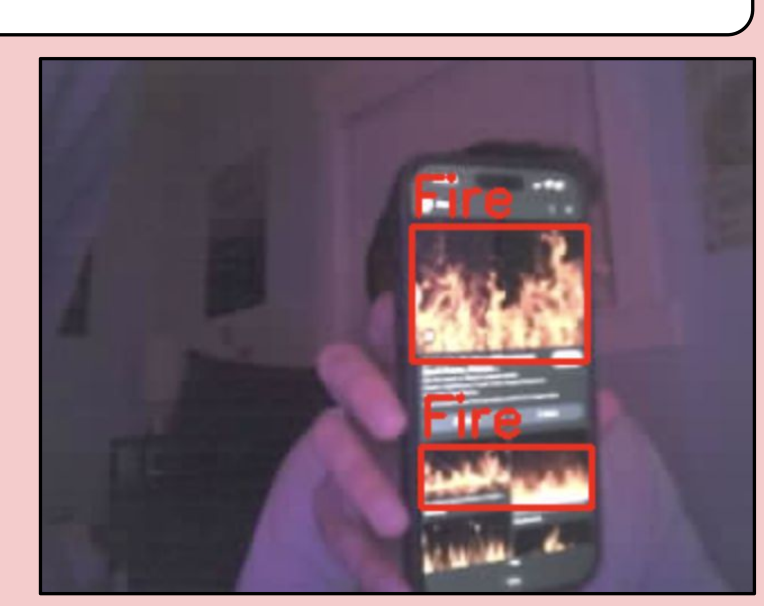
DEMO AND VALIDATION



Ground Station Map
- Drone telemetry and estimated fire coordinates are displayed in real time

- End-to-end telemetry pipeline tested with simulated drone nodes
- Fire alerts displayed on live map interface
- Triangulation logic validated with unit tests



Fire Detection View
- YOLOv8 identifies sustained fire/smoke from the camera stream